

6 Empirical Forecasting Design

This chapter defines the empirical design used to forecast hourly grid load in Germany and Austria. The methods introduced in Chapter 3 are applied to the processed data from Chapter 4, while the findings of Chapter 5 guide the choice of model structures and explanatory variables. The chapter specifies the forecasting task, information set, data partitions, candidate models, rolling procedure and evaluation framework. The results are presented in Chapter 7.

6.1 Forecasting Objective and Comparison Framework

The goal is to forecast the hourly grid-load level for the next day. Germany and Austria are modelled separately because their load levels and temporal patterns differ. Each model produces one forecast for every hourly interval of the target day. The archived official SMARD day ahead forecast is used as the main operational benchmark, allowing the statistical models to be compared with both simple baselines and a forecast used in practice. APG is not used as a separate benchmark because its Austrian forecast values are identical to those published by SMARD over their common period.

The analysis does not attempt to reproduce the forecasting systems used by the transmission system operators, whose complete inputs and procedures are unavailable but instead it examines how accurately transparent classical time-series models can forecast hourly day-ahead grid load and how their performance compares with SMARD when evaluated on the same target intervals.

6.2 Forecast Origins and Information Sets

A clear forecast origin is required to ensure that no future information enters the models. For each German target day, the forecast origin is set at 18:00 on the preceding day, following the SMARD submission schedule. For Austria, it is set at 08:00 on the preceding day, corresponding to the APG publication schedule (Austrian Power Grid AG, 2026; SMARD, 2026).

Because the forecast is made at different times in the two countries, the time between the forecast origin and the start of the target day also differs. For Germany, this interval is approximately 6 hours, while for Austria it is approximately 16 hours. For the final hour of the target day, the corresponding lead times are approximately 29 hours and 39 hours. These differences result from the official publication schedules. The model forecasts are therefore compared with the SMARD forecasts separately within each country.

Only load observations with an end time no later than the forecast origin are treated as available. Calendar variables for the target day, including the hour, weekday, month and public holidays, may be used because they are known in advance. Future actual load and observed temperature are excluded. Observed temperature would not have been known at the forecast origin.

A target day normally contains 24 hourly intervals. Days affected by the transition to or from daylight-saving time contain 23 or 25 intervals. All existing intervals are forecast and matched using their complete timestamps without forcing these days into a 24-hour structure.

6.3 Training, Validation and Test Design

The data are divided chronologically to reflect how forecasts are produced in practice. Random splitting is unsuitable because it could allow later observations to influence models used to predict earlier periods. The same calendar periods are used for Germany and Austria, although the models for the two countries are estimated and selected separately.

Data from 2020 to 2023 are used for initial model estimation. Forecasts for 2024 form the validation period and are used to compare candidate specifications, select predictors and determine the final model settings. Each validation forecast is generated using only the observations available at its forecast origin and follows the rolling procedure described in Section 6.4.

After the model choices have been fixed, the selected models are re-estimated using all observations available at the first forecast origin of 2025. During the test period, forecasts are produced sequentially. The estimation window may be expanded with new actual observations only after those observations have become available at the relevant forecast origin. Consequently, earlier observations from 2025 may be used when producing later forecasts, but no observation is used before it becomes available.

The archived SMARD forecasts are not used to estimate or select the statistical models. They remain an external operational benchmark against which the final model forecasts are evaluated..

6.4 Rolling Day-Ahead Forecasting Procedure

Forecasts are generated using a rolling-origin procedure. At each country-specific forecast origin, every method uses only the observations available at that time. Methods that require a multi-step forecast generate a path from the first unavailable interval to the end of the target day. Any intermediate forecasts are needed for the calculation but are not evaluated. If an unavailable lagged value is required, its forecast is used instead of the later actual value.

Each day-ahead forecast is fixed at its origin and is not revised when new observations become available. Before the next forecast is produced, newly available actual values are added to the expanding estimation window. The frequency of complete model re-estimation is selected during validation and remains unchanged during testing.

Figure 22 summarizes the rolling day-ahead forecasting and evaluation procedure used for both countries.

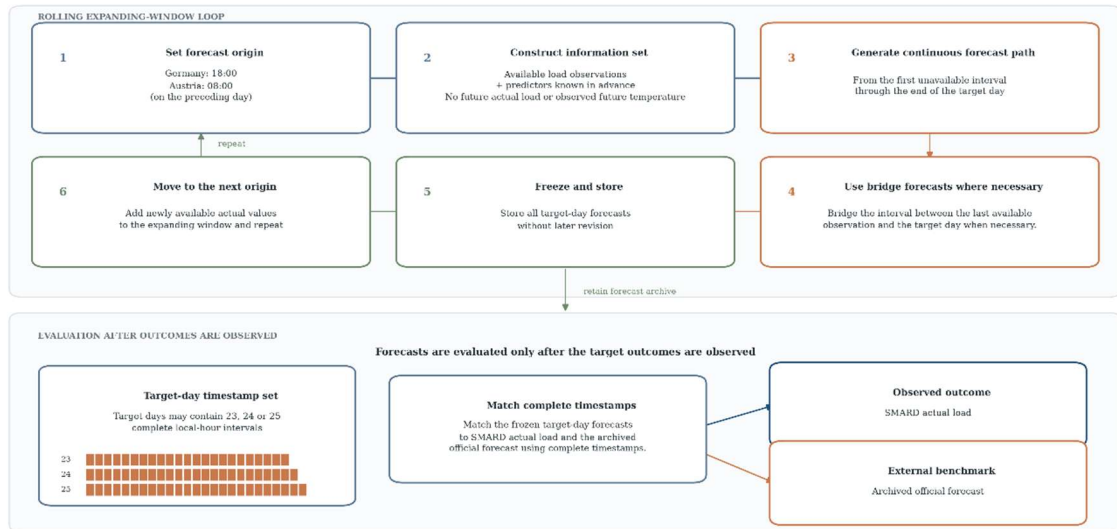


Figure 22: Rolling day-ahead forecasting procedure. Own illustration.

Each forecast is stored with its country, method, forecast origin and target timestamp. It is then matched with the actual load and official SMARD forecast for the same interval. The same rolling procedure is used throughout the 2024 validation period and the 2025 test period.

6.5 Candidate Forecasting Models

The study compares simple baselines, univariate time-series models and models with external variables. Each model group is applied separately to Germany and Austria, with final specifications selected using the 2024 validation period. The mathematical foundations were presented in Chapter 3.

6.5.1 Baseline Models

Four baseline methods are included. The naïve method uses the latest available load value for all future hours. The daily seasonal-naïve method uses the corresponding hour from the preceding day. If that value is unavailable at the forecast origin, the forecast generated earlier in the forecast path is used. The weekly seasonal-naïve method uses the corresponding hour from the previous week.

The average hour-of-week profile calculates a historical mean for each weekday-hour combination. All averages use only observations available at the forecast origin. Seasonal matching is based on local calendar time, while complete timestamps are retained during daylight-saving-time changes. These baselines provide reference levels for evaluating the more complex models.

6.5.2 Exponential Smoothing Models

Additive and multiplicative Holt-Winters models are considered. Daily and weekly seasonal periods are examined separately, while trend and damped-trend variants are

compared during validation. Simple exponential smoothing and non-seasonal Holt models are excluded because they cannot represent the strong seasonal patterns identified in Chapter 5.

6.5.3 ARIMA and SARIMA Models

A non-seasonal ARIMA model is included as a reference for short-term dependence. The SARIMA models use a daily seasonal period of 24 hours. Weekly effects are represented by the weekly baseline and by calendar predictors in models with external variables. Candidate orders are identified using the training data.

6.5.4 Regression with ARIMA Errors and SARIMAX

The final model group combines predictors known at the forecast origin with an ARIMA or SARIMA error structure. Non-seasonal variants use ARIMA errors. Seasonal variants use SARIMA errors and are referred to as SARIMAX models. The predictors include calendar variables and public-holiday indicators. Temperature is included only if suitable archived day-ahead forecasts are available. These models are compared with the corresponding univariate models to assess the value of the additional information.

6.6 Predictor Variables and Their Availability

Predictors in the regression with ARIMA errors and SARIMAX models are based on the patterns identified in Chapter 5. Only information known at the forecast origin is used.

The calendar predictors include hour of day, day of week, month and a country-specific public-holiday indicator. They are created from the target timestamp and represented by dummy variables. When an intercept is included, one category from each group is omitted as the reference. A separate weekend indicator is unnecessary because weekends are already represented by the day-of-week variables.

Holiday indicators follow the country-specific definitions established in Chapter 4. Regional German holidays are not classified as nationwide events. The same predictor definitions are used during training, validation and testing.

The candidate model families, seasonal structures and predictor sets are summarized in **Table 4**.

Table 4: Candidate models and predictor sets.

Model family	Candidate variants	Seasonal structure	Predictor set
Naïve	Last available observation	None	Lagged target only

Daily seasonal naïve	Fixed 24-hour lag	Daily (24 hours)	Lagged target only
Weekly seasonal naïve	Fixed 168-hour lag	Weekly (168 hours)	Lagged target only
Hour-of-week profile	Historical hour-of-week profile	Weekly hour-of-week profile	Historical target profile
Holt-Winters	Additive and multiplicative; trend and damped-trend variants	Daily (24-hour) and weekly (168-hour) seasonality	Lagged target only
ARIMA	Order (p,d,q) selected using training and validation data	None	Lagged target only
SARIMA	Candidate non-seasonal and seasonal orders	Seasonal period is 24 hours. Weekly effects are represented through weekly baselines or calendar predictors rather than a seasonal period of 168.	Lagged target only
Regression with ARIMA errors	Calendar-predictor regression with ARIMA errors	Model-dependent	Hour of day; day of week; month; country-specific public holidays. No separate weekend dummy when day-of-week variables are used.
SARIMA X	Calendar-predictor regression with	Model-dependent	Hour of day; day of week; month; country-specific public holidays. No separate weekend dummy

	seasonal ARIMA errors		when day-of-week variables are used.
Note: Temperature is considered only as an optional extension if genuine archived day-ahead temperature forecasts are available.			

The temperature data examined in Chapter 5 contain observed values that would not have been available when the forecasts were produced. Temperature is therefore excluded from the main experiment. It may be examined separately only if suitable archived day-ahead temperature forecasts are available.

6.7 Model Estimation, Selection and Diagnostics

Initial model specifications are developed separately for Germany and Austria using the 2020–2023 training period. AIC, AICc and BIC are used to compare specifications within the same model family and fitted to the same observations. For ARIMA-based models, information criteria are compared only between specifications with the same differencing orders. These criteria narrow the candidate set but do not determine the final model alone.

The remaining specifications are evaluated through rolling day-ahead forecasts for 2024. MAE across all target intervals is the primary selection criterion. RMSE and MAPE provide additional information about large errors and relative accuracy. Model orders, seasonal structures, predictor sets and updating procedures are selected separately for each country.

Residual plots, the ACF and the Ljung–Box test are used to check for systematic bias and remaining autocorrelation. The AR and MA components are also checked for stationarity and invertibility. Models with convergence problems, unstable estimates or clear residual patterns are rejected even if they produce a small improvement in validation accuracy.

After selection, each final model is re-estimated using all observations available before the first forecast origin of 2025. The models are then evaluated in 2025 without further changes to their specifications.

6.8 Forecast Evaluation Framework

Forecast accuracy is evaluated using the 2025 test period. For each target interval i , the forecast error is

$$e_i = y_i - \hat{y}_i$$

where y_i is the actual load and $\hat{y}_i$ is the forecast. The accuracy measures are

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |e_i|$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n e_i^2}$$

and

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{e_i}{y_i} \right|$$

where n is the number of evaluated intervals. MAE is the primary ranking measure, RMSE gives greater weight to large errors and MAPE expresses errors relative to actual load. MAPE is suitable because national grid load remains above zero.

All methods, including the official SMARD forecast, are evaluated against the same actual values and common target timestamps. Missing forecasts are not imputed, and forecast coverage is reported separately. Accuracy is calculated for the complete test period and by target hour, season, weekday or weekend status, and public-holiday status. Holiday results are interpreted cautiously because they contain fewer observations.

Germany and Austria are evaluated separately. MAE and RMSE are not compared directly because their load levels differ. MAPE allows a relative comparison, although this remains descriptive because the countries have different forecast origins and information sets. The results are presented in Chapter 7.

6.9 Implementation and Reproducibility

The forecasting procedure will be implemented in the same 64-bit CPython 3.12.10 environment used for data preparation and exploratory analysis. pandas and NumPy will be used for data handling and accuracy calculations, statsmodels for forecasting and residual diagnostics, and Matplotlib for figures. Exact package versions will be recorded in the project requirements and lock files.

The pipeline will use the processed hourly datasets described in Chapter 4 without changing the raw source files. Forecast origins, data partitions, model specifications, predictors and updating schedules will be defined in the code. Forecasts will be saved before the corresponding actual load and official SMARD forecast are added for evaluation. Validation scores, selected specifications, residual diagnostics and estimation warnings will also be retained.

Automated tests will check for duplicated timestamps, missing intervals, incorrect forecast origins, future-information leakage, incomplete forecast coverage and daylight-saving-time errors. The complete test suite and dependency checks will be run before the results are produced. This creates a clear link between each result in Chapter 7 and its data, model specification and forecast origin.

7 Forecasting Results

This chapter presents the empirical forecasting results for Germany and Austria. It first reports the validation-based model selection and final model specifications, followed by the 2025 out-of-sample accuracy results and comparison with the SMARD forecasts. The analysis then examines performance across hours and calendar conditions, the contribution of model components and predictors, representative forecast profiles, and residual diagnostics.

- 7.1 Validation Results and Final Model Specifications**
- 7.2 Overall Out-of-Sample Accuracy and SMARD Comparison**
 - 7.2.1 Germany**
 - 7.2.2 Austria**
 - 7.2.3 Cross-Country Comparison**
- 7.3 Accuracy across Hours and Calendar Conditions**
 - 7.3.1 Accuracy by Hour of Day**
 - 7.3.2 Accuracy by Season**
 - 7.3.3 Accuracy by Day Type**
- 7.4 Contribution of Model Components and Predictor Variables**
- 7.5 Representative Day-Ahead Forecast Profiles**
- 7.6 Residual Diagnostics and Robustness Checks**
- 7.7 Summary of Forecasting Results**

8 Discussion

8.1 Interpretation of the Main Forecasting Results

8.2 Influence of Seasonality, Calendar Effects and Temperature

8.3 Demand Changes and Forecasting Performance during Crisis Periods

8.4 Differences between Austria and Germany

8.5 Comparison with Official Forecasts

8.6 Practical Relevance of the Results

8.7 Limitations of the study

9 Conclusion and Future Work

9.1 Summary of the Thesis

9.2 Main Findings and Answers to the Research Questions

9.3 Scientific and Practical Contribution

9.4 Recommendations for Future Research

10 Appendix

10.1 Additional tables

10.2 Additional plots

10.3 Data preprocessing details

10.4 Model parameter tables

10.5 Python code snippets

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Appendix

